EarlyCare AI | Lean Canvas

An AI agent for early detection of chronic diseases using symptom checks and machine learning

SDG 3: Good Health and Well-being

IBM SkillsBuild Masterclass 5 Project

1 PROBLEM

- **Late detection:** millions of Indians live with diabetes, heart disease, hypertension and kidney disease without knowing it.
- **Unscreened prediabetes:** 136 M Indians have prediabetes, a window where disease can be prevented.
- **Access gap:** ~80% specialist shortfall at rural CHCs; a check-up can cost a day's wages.
- **Barriers:** English-only apps, low literacy, and no follow-up after one-time camps.

EXISTING ALTERNATIVES

- Google / global symptom-checker apps
- Paid full-body check-up packages
- Government NCD screening camps
- Waiting for symptoms to appear

2 SOLUTION

- **Talk:** chat or voice in Kannada, Hindi and English (WhatsApp bot / Android app).
- **Assess:** ML models (scikit-learn / XGBoost) score risk for diabetes, heart disease, high BP and kidney disease, with SHAP reasons.
- **Act:** nearest PHC or lab, teleconsult links, follow-up reminders; ASHA worker mode for households.

8 KEY METRICS

- People screened (1,000 pilot to 1 M by Year 3)
- Follow-up test rate (50% of high-risk users in 30 days)
- Model recall 85%+ on validation data
- Cost per screening under Rs 10

3 UNIQUE VALUE PROPOSITION

- **Know your risk in 3 minutes, in your own language, for free.**
- A health-check camp in your pocket: a multilingual AI agent that estimates risk of four chronic diseases and tells you exactly what to do next.
- Screening and guidance, never a diagnosis.

HIGH-LEVEL CONCEPT

- A symptom-checker app + a rural health camp + a follow-up reminder service, in one WhatsApp chat.

5 UNFAIR ADVANTAGE

- Rural-first design: voice and local languages
- PHC and ASHA worker partnerships
- India-specific data and Bhashini language support
- Clinical advisor-reviewed questions and red-flag rules

4 CHANNELS

- ASHA workers and ANMs
- NGO / NSS health camps
- WhatsApp bot
- Google Play Store

6 CUSTOMER SEGMENTS

- **Individuals:** adults 30+ in Tier-2/3 towns and villages.
- **ASHA & ANM workers:** frontline household screeners.
- **PHCs & health missions:** public health centres, state programmes.
- **Employers & insurers:** wellness programmes.

EARLY ADOPTERS

- Smartphone users aged 30-60 who already worry about sugar or BP
- ASHA workers in one Karnataka pilot district
- NGOs and NSS units running free camps

7 COST STRUCTURE

- **Cloud & AI:** IBM watsonx.ai (Granite), Cloud Code Engine, Cloudant, Bhashini and WhatsApp Business API usage.
- **Development:** ML model training and validation, chatbot, Android app, PHC/ASHA dashboard.
- **Clinical review:** clinical advisor for questions, thresholds and red-flag rules.
- **Outreach:** ASHA onboarding, camp partnerships, local-language content.

9 REVENUE STREAMS

- **Free for individuals:** core screening is always free.
- **B2G licences:** per-screening fee or annual licence for state health missions and PHCs.
- **B2B wellness plans:** per-employee pricing for employers and insurers.
- **Lab referrals:** fee from partner labs for confirmatory tests booked via the app.
- **CSR & grants:** health-tech CSR funds and innovation grants for rural roll-out.

Sources: World Bank; ICMR-INDIAB 2023; Health Dynamics of India 2022-23, MoHFW